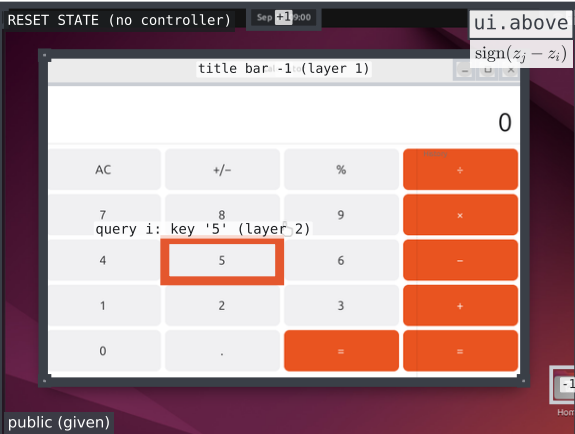
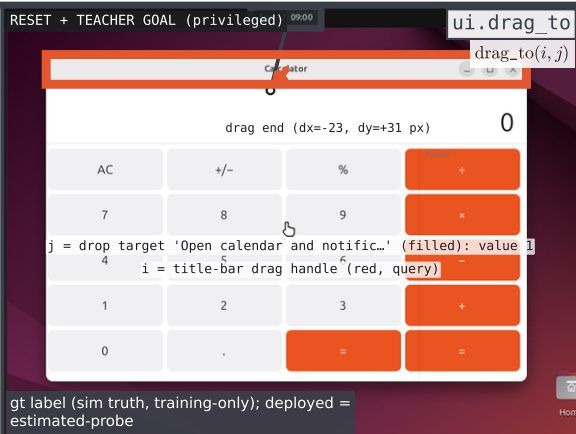


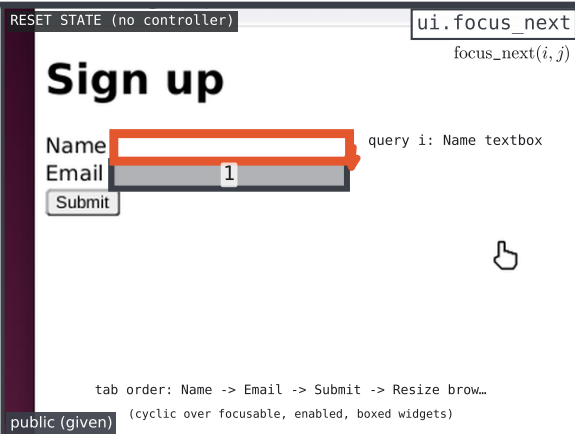
UI (5 implemented)



ui.above: signed render order: +1 if widget j is drawn above i (higher dense z-layer), -1 below, 0 same layer; dark = +1 (resize handles, shell), light = -1 (window frame, desktop icon), op order over field zlayer, form bias (antisymmetric). Source: public (given widget tree).



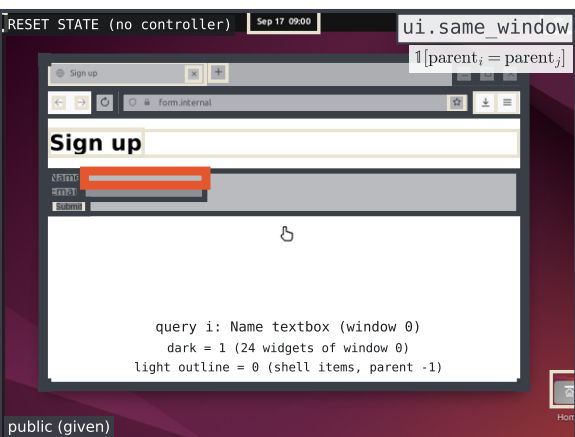
ui.drag_to: pair label: the teacher's drag handle -> the nearest widget outside the dragged window to where the drag ends (cw/drag_window); 1 pair of 2116 valid, op bilinear, form aug (the bias IS the pair probe). Source: gt label (teacher goal), training-only; deployed = estimated-probe.



ui.focus_next: public UI edge: i, j consecutive in the tab order (focusable, enabled widgets in scene order, cyclic); Name -> Email, op edge, form bias (edges:ui-rel-v1 channel focus_next). Source: public (given widget tree).



ui.label_for: public UI edge: a role=label widget -> the next widget after it in the same window (scene order); here 'Email' -> its textbox, op edge, form bias (edges:ui-rel-v1 channel label_for). Source: public (given widget tree).



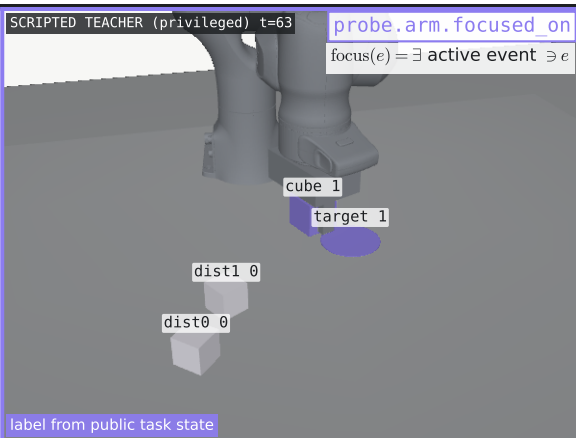
ui.same_window: 1 when two widgets belong to the same window (parent_id equality; shell items such as the tab strip have parent -1 and never match), op same over field parent_id, form bias. Source: public (given widget tree).

PROBE READOUTS (26 implemented)

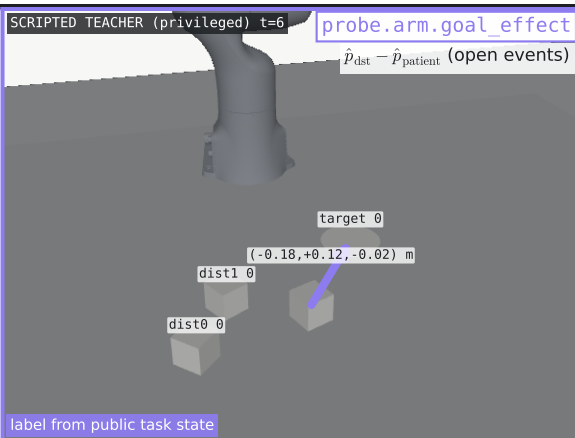
tile colour = what the readout encodes: cyan geometric quantity, amber physical interaction, violet task / procedure state



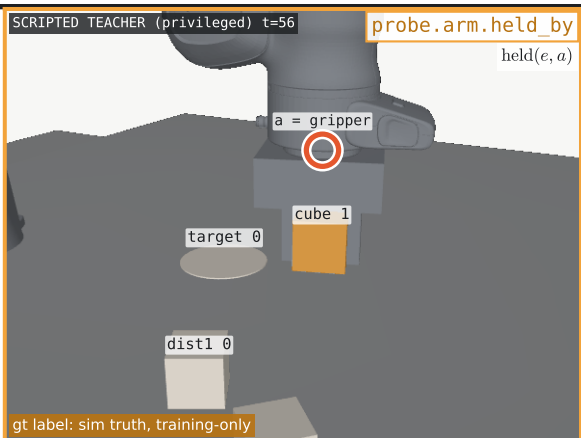
probe.arm.acting_on: readout of whether assembly a's hand links touch entity e (contact truth), readout (inert), address entity x assembly, bce. Source: gt sim truth (Session.truth held_by / contacts), training-only.



probe.arm.focused_on: readout of which entities a currently ACTIVE task event binds (active op 'place': cube + target), readout (inert), address entity, bce. Source: public label (task runtime status + bindings).



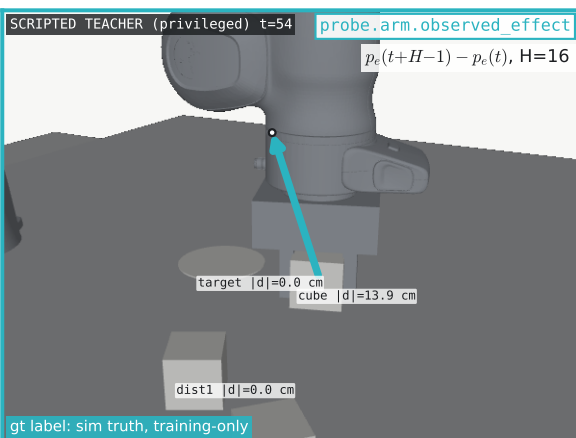
probe.arm.goal_effect: readout of the displacement the task asks of each entity: destination minus patient for open events (cube -> target; others 0), readout (inert), address entity, gauss. Source: public label (task spec + public pose estimates).



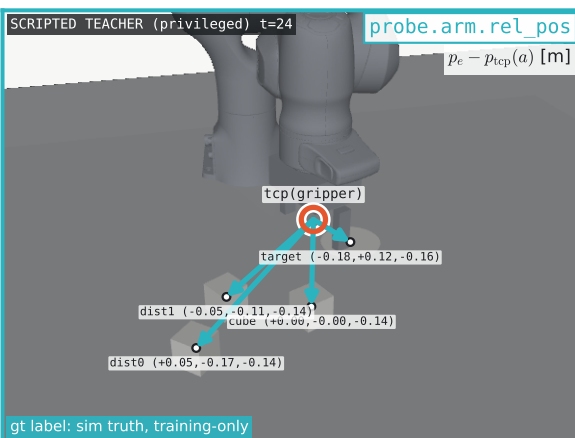
probe.arm.held_by: readout of whether entity e is held by assembly a (here gripper; cube 1, others 0), readout (inert), address entity x assembly, bce. Source: gt sim truth (Session.truth held_by / contacts), training-only.



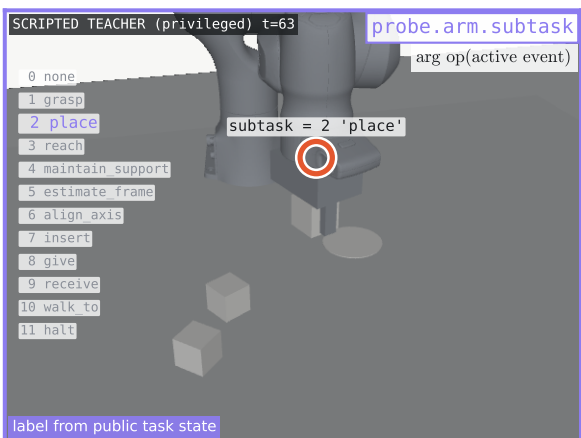
probe.arm.looking_at: readout of each entity's angle off the front-camera optical axis (gaze label, degrees, scaled 1/30), readout (inert) on the packet, address entity, mse. Source: gt sim truth (camera pose + body pose), training-only.



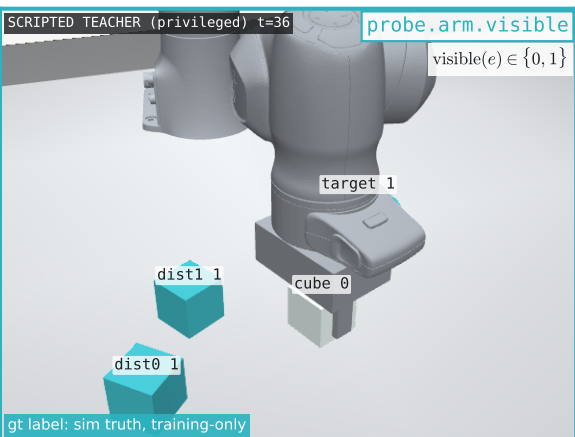
probe.arm.observed_effect: readout of each entity's realized displacement over the 16-tick chunk (future_disp; cube lifted, others 0), readout (inert), address entity, gauss. Source: gt sim truth (future state), training-only.



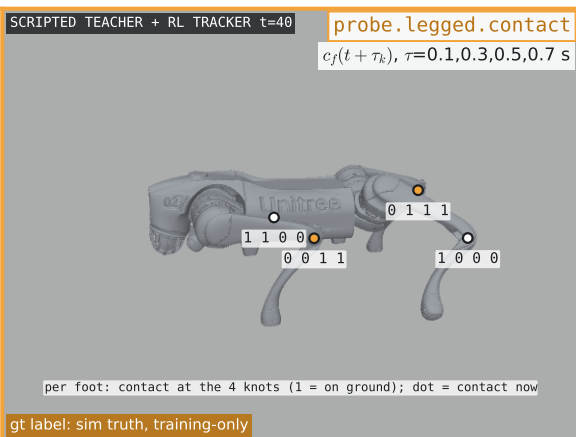
probe.arm.rel_pos: readout of entity position relative to the assembly's TCP, (x,y,z) m, world frame, readout (inert), address entity x assembly, gauss (mu, logvar). Source: gt sim truth, training-only.



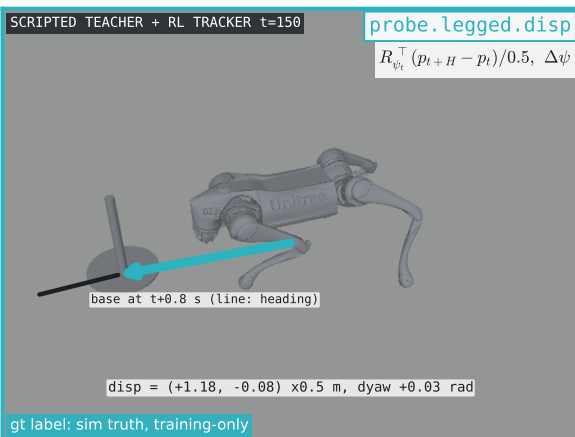
probe.arm.subtask: readout of the active task operator for the assembly (12-way; this episode: grasp -> place -> none), readout (inert), address assembly, ce. Source: public label (task runtime: first ACTIVE event's operator).



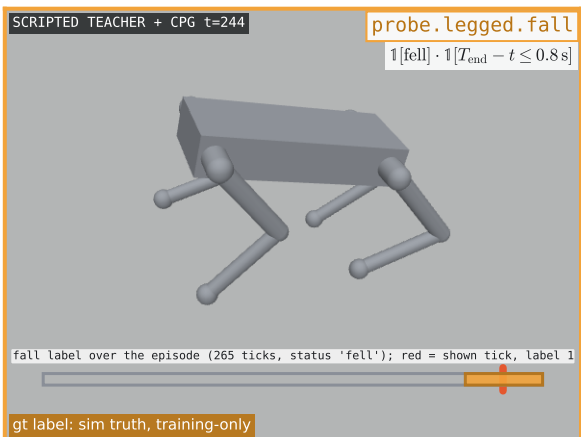
probe.arm.visible: readout of whether each entity is visible in the front camera (cube occluded by the closing gripper here: 0), readout (inert) on the packet, address entity, bce. Source: gt sim truth, training-only target; deployed = probe estimate.



probe.legged.contact: readout of each foot's ground contact at the 4 packet knots (+0.1/0.3/0.5/0.7 s) of a trot, readout (inert), address knot x assembly (feet), bce. Source: gt sim truth (future contacts), training-only.



probe.legged.disp: readout of the base displacement over the next 0.8 s in the start body frame (xy in 0.5 m units, yaw change in rad), readout (inert), address assembly, gauss (3-d). Source: gt sim truth (future base pose), training-only.



probe.legged.fall: readout of an imminent fall: 1 in the last 0.8 s of an episode that ends in a fall (non-foot ground contact, low base or tilt), readout (inert), address assembly, bce. Source: gt sim truth (episode outcome), training-only.